

Table 1: Computational comparison for generalized linear models using Python with a dense design matrix, $n = 2,000$ and $p = 1,001$ (including the intercept). Runtime is reported in seconds. “Iteration” denotes the number of outer iterations, “Rel. grad.” denotes the relative gradient norm at termination, and “Converged” reports whether the solver declared convergence.

Model	Method	Time (s)	Iteration	Rel. grad.	Converged
Gaussian-identity	Spectral-MM	0.048	25	9.38×10^{-9}	Yes
	CRLS	0.031	2	7.38×10^{-9}	Yes
	LSMR	0.031	2	7.38×10^{-9}	Yes
	CG	0.032	2	7.34×10^{-9}	Yes
	CGLS	0.032	2	7.34×10^{-9}	Yes
	LSQR	0.033	2	7.34×10^{-9}	Yes
	<code>statsmodels.GLM</code>	0.515	3	5.12×10^{-15}	Yes
Bernoulli-logit [‡]	Spectral-MM	0.208	54	6.21×10^{-9}	Yes
	CRLS	0.612	11	5.29×10^{-9}	Yes
	LSMR	0.613	11	5.29×10^{-9}	Yes
	CG	0.623	11	5.28×10^{-9}	Yes
	CGLS	0.625	11	5.32×10^{-9}	Yes
	LSQR	0.628	11	5.27×10^{-9}	Yes
	<code>statsmodels.GLM</code>	3.557	42	5.08×10^{-16}	Yes
Bernoulli-probit [‡]	Spectral-MM	0.152	37	6.68×10^{-9}	Yes
	CRLS	0.618	11	2.91×10^{-10}	Yes
	LSMR	0.621	11	2.91×10^{-10}	Yes
	CG	0.631	11	2.83×10^{-10}	Yes
	LSQR	0.632	11	2.82×10^{-10}	Yes
	CGLS	0.632	11	2.81×10^{-10}	Yes
	<code>statsmodels.GLM</code>	3.332	39	2.73×10^{-15}	Yes
Poisson-log	Spectral-MM	0.108	34	1.04×10^{-8}	Yes [†]
	CRLS	0.135	6	5.66×10^{-10}	Yes
	LSMR	0.138	6	5.66×10^{-10}	Yes
	CG	0.147	6	5.28×10^{-10}	Yes
	LSQR	0.149	6	6.08×10^{-10}	Yes
	CGLS	0.150	6	6.05×10^{-10}	Yes
	<code>statsmodels.GLM</code>	1.135	11	1.11×10^{-14}	Yes
Gamma-log	Spectral-MM	0.085	59	2.68×10^{-8}	Yes [†]
	CG	0.699	45	7.50×10^{-9}	Yes
	CRLS	0.702	49	5.41×10^{-9}	Yes
	CGLS	0.708	45	7.50×10^{-9}	Yes
	LSQR	0.711	45	7.50×10^{-9}	Yes
	LSMR	0.714	49	5.41×10^{-9}	Yes
	<code>statsmodels.GLM</code>	89.726	1000	2.41×10^{-2}	No
Negative-binomial-log	Spectral-MM	0.079	39	2.39×10^{-8}	Yes [†]
	CRLS	0.439	28	7.34×10^{-9}	Yes
	LSMR	0.447	28	7.34×10^{-9}	Yes
	CG	0.488	28	4.98×10^{-9}	Yes
	LSQR	0.496	28	4.98×10^{-9}	Yes
	CGLS	0.496	28	4.98×10^{-9}	Yes
	<code>statsmodels.GLM</code>	10.308	21	5.30×10^{-9}	Yes

[†]The solver declared convergence, but neither $\|g\| \leq 10^{-6}$ nor $\|g\|/(1 + \|g_0\|) \leq 10^{-8}$ was satisfied. [‡]These data exhibit separation/perfect prediction; a small gradient does not establish a finite maximum-likelihood estimate. Spectral-MM and the Krylov methods use the shared C++ core. Native timings are medians of five interleaved rounds after warm-up. GLM timings are medians of three runs from an earlier timing session; cross-session timing variation was observed. Sparse GLM inputs were converted to dense, with conversion included in timing. The negative-binomial shape parameter is fixed at $\theta = 4$. All native/Julia fits use $\rho = 10^{-6}$ and no ridge penalty.

Table 2: Computational comparison for generalized linear models using Python with a sparse design matrix, $n = 2,000$ and $p = 1,001$ (including the intercept). The non-intercept design has approximately 20 nonzeros per row (about 2% density). Runtime is reported in seconds. “Iteration” denotes the number of outer iterations, “Rel. grad.” denotes the relative gradient norm at termination, and “Converged” reports whether the solver declared convergence.

Model	Method	Time (s)	Iteration	Rel. grad.	Converged
Gaussian-identity	Spectral-MM	0.005	25	1.19×10^{-8}	Yes [†]
	CG	0.002	2	6.76×10^{-9}	Yes
	CGLS	0.002	2	6.76×10^{-9}	Yes
	CRLS	0.002	2	6.81×10^{-9}	Yes
	LSQR	0.002	2	6.76×10^{-9}	Yes
	LSMR	0.002	2	6.81×10^{-9}	Yes
	statsmodels.GLM	0.510	3	1.35×10^{-14}	Yes
Bernoulli-logit [‡]	Spectral-MM	0.053	86	5.12×10^{-9}	Yes
	CRLS	0.102	16	1.34×10^{-9}	Yes
	CGLS	0.103	16	2.00×10^{-9}	Yes
	CG	0.103	16	2.59×10^{-9}	Yes
	LSQR	0.104	16	2.93×10^{-9}	Yes
	LSMR	0.105	16	1.34×10^{-9}	Yes
	statsmodels.GLM	20.618	47	1.44×10^{-15}	Yes
Bernoulli-probit [‡]	Spectral-MM	0.052	76	9.07×10^{-9}	Yes
	CG	0.089	14	7.70×10^{-9}	Yes
	CGLS	0.090	14	7.94×10^{-9}	Yes
	LSQR	0.092	14	7.66×10^{-9}	Yes
	CRLS	0.095	16	5.48×10^{-9}	Yes
	LSMR	0.098	16	5.14×10^{-9}	Yes
	statsmodels.GLM	3.575	42	1.09×10^{-14}	Yes
Poisson-log	Spectral-MM	0.007	30	5.85×10^{-9}	Yes
	CRLS	0.006	5	9.87×10^{-9}	Yes
	CG	0.006	5	9.91×10^{-9}	Yes
	LSMR	0.006	5	9.87×10^{-9}	Yes
	CGLS	0.006	5	9.91×10^{-9}	Yes
	LSQR	0.006	5	9.91×10^{-9}	Yes
	statsmodels.GLM	0.889	7	7.84×10^{-15}	Yes
Gamma-log	Spectral-MM	0.017	78	3.43×10^{-8}	Yes [†]
	CRLS	0.036	38	1.69×10^{-8}	Yes [†]
	LSMR	0.039	38	1.69×10^{-8}	Yes [†]
	CG	0.039	40	9.41×10^{-9}	Yes
	CGLS	0.041	41	5.05×10^{-9}	Yes
	LSQR	0.041	40	9.41×10^{-9}	Yes
	statsmodels.GLM	306.984	1000	8.25×10^{-2}	No
Negative-binomial-log	Spectral-MM	0.009	34	1.72×10^{-8}	Yes [†]
	CRLS	0.017	16	1.75×10^{-8}	Yes [†]
	LSMR	0.018	16	1.75×10^{-8}	Yes [†]
	CG	0.018	17	6.16×10^{-9}	Yes
	CGLS	0.019	17	6.16×10^{-9}	Yes
	LSQR	0.019	17	6.16×10^{-9}	Yes
	statsmodels.GLM	7.910	16	1.16×10^{-8}	Yes [†]

[†]The solver declared convergence, but neither $\|g\| \leq 10^{-6}$ nor $\|g\|/(1 + \|g_0\|) \leq 10^{-8}$ was satisfied. [‡]These data exhibit separation/perfect prediction; a small gradient does not establish a finite maximum-likelihood estimate. Spectral-MM and the Krylov methods use the shared C++ core. Native timings are medians of five interleaved rounds after warm-up. GLM timings are medians of three runs from an earlier timing session; cross-session timing variation was observed. Sparse GLM inputs were converted to dense, with conversion included in timing. The negative-binomial shape parameter is fixed at $\theta = 4$. All native/Julia fits use $\rho = 10^{-6}$ and no ridge penalty.

Table 3: Computational comparison for non-GLM models using Python with a dense design matrix, $n = 2,000$ and $p = 1,001$ (including the intercept). Runtime is reported in seconds. “Iteration” denotes the number of outer iterations, “Rel. grad.” denotes the relative gradient norm at termination, and “Converged” reports whether the solver declared convergence.

Model	Method	Time (s)	Iteration	Rel. grad.	Converged
Smooth quantile	Spectral-MM	0.236	40	2.74×10^{-8}	Yes [†]
	CRLS	1.107	21	7.87×10^{-10}	Yes
	CGLS	1.171	20	7.95×10^{-10}	Yes
	LSQR	1.173	20	8.19×10^{-10}	Yes
	LSMR	1.243	23	6.29×10^{-9}	Yes
	CG	1.713	29	8.31×10^{-10}	Yes
Expective	Spectral-MM	0.062	28	1.15×10^{-8}	Yes [†]
	CRLS	0.082	5	3.47×10^{-10}	Yes
	LSMR	0.083	5	3.47×10^{-10}	Yes
	CG	0.088	5	3.59×10^{-10}	Yes
	LSQR	0.089	5	3.59×10^{-10}	Yes
	CGLS	0.089	5	3.59×10^{-10}	Yes
Pseudo-Huber	Spectral-MM	0.068	30	7.55×10^{-9}	Yes
	CRLS	0.175	9	3.79×10^{-9}	Yes
	LSMR	0.177	9	3.79×10^{-9}	Yes
	CG	0.190	9	3.69×10^{-9}	Yes
	CGLS	0.190	9	3.69×10^{-9}	Yes
	LSQR	0.191	9	3.69×10^{-9}	Yes
Student- t	Spectral-MM	0.076	43	5.28×10^{-8}	Yes [†]
	CRLS	0.353	25	7.14×10^{-9}	Yes
	LSMR	0.360	25	6.20×10^{-9}	Yes
	CG	0.370	24	8.54×10^{-9}	Yes
	LSQR	0.374	24	8.54×10^{-9}	Yes
	CGLS	0.374	24	8.54×10^{-9}	Yes

[†]The solver declared convergence, but neither $\|g\| \leq 10^{-6}$ nor $\|g\|/(1 + \|g_0\|) \leq 10^{-8}$ was satisfied. Spectral-MM and the Krylov methods use the shared C++ core. Native timings are medians of five interleaved rounds after warm-up. Parameters: smooth quantile $(\tau, \epsilon) = (0.25, 0.1)$; expectile $\tau = 0.25$; pseudo-Huber $\delta = 1$; Student- t $(\nu, \sigma) = (4, 1)$. All native/Julia fits use $\rho = 10^{-6}$ and no ridge penalty.

Table 4: Computational comparison for non-GLM models using Python with a sparse design matrix, $n = 2,000$ and $p = 1,001$ (including the intercept). The non-intercept design has approximately 20 nonzeros per row (about 2% density). Runtime is reported in seconds. “Iteration” denotes the number of outer iterations, “Rel. grad.” denotes the relative gradient norm at termination, and “Converged” reports whether the solver declared convergence.

Model	Method	Time (s)	Iteration	Rel. grad.	Converged
Smooth quantile	Spectral-MM	0.021	43	9.25×10^{-9}	Yes
	CRLS	0.079	19	1.82×10^{-9}	Yes
	LSMR	0.082	19	1.15×10^{-9}	Yes
	CG	0.089	19	3.10×10^{-9}	Yes
	CGLS	0.090	19	5.30×10^{-9}	Yes
	LSQR	0.092	19	4.68×10^{-9}	Yes
Expective	Spectral-MM	0.006	25	9.93×10^{-9}	Yes
	CRLS	0.006	5	1.32×10^{-10}	Yes
	CG	0.006	5	1.11×10^{-10}	Yes
	CGLS	0.006	5	1.11×10^{-10}	Yes
	LSMR	0.006	5	1.32×10^{-10}	Yes
	LSQR	0.006	5	1.11×10^{-10}	Yes
Pseudo-Huber	Spectral-MM	0.008	26	6.17×10^{-9}	Yes
	CRLS	0.013	10	4.07×10^{-9}	Yes
	CG	0.014	10	4.04×10^{-9}	Yes
	LSMR	0.014	10	4.07×10^{-9}	Yes
	CGLS	0.014	10	4.04×10^{-9}	Yes
	LSQR	0.015	10	4.04×10^{-9}	Yes
Student- t	Spectral-MM	0.010	42	8.82×10^{-9}	Yes
	CRLS	0.025	25	6.23×10^{-9}	Yes
	CG	0.026	25	7.02×10^{-9}	Yes
	LSMR	0.027	25	6.23×10^{-9}	Yes
	CGLS	0.027	25	7.02×10^{-9}	Yes
	LSQR	0.027	25	7.02×10^{-9}	Yes

[†]The solver declared convergence, but neither $\|g\| \leq 10^{-6}$ nor $\|g\|/(1 + \|g_0\|) \leq 10^{-8}$ was satisfied. Spectral-MM and the Krylov methods use the shared C++ core. Native timings are medians of five interleaved rounds after warm-up. Parameters: smooth quantile $(\tau, \epsilon) = (0.25, 0.1)$; expectile $\tau = 0.25$; pseudo-Huber $\delta = 1$; Student- t $(\nu, \sigma) = (4, 1)$. All native/Julia fits use $\rho = 10^{-6}$ and no ridge penalty.

Table 5: Computational comparison for generalized linear models using R with a dense design matrix, $n = 2,000$ and $p = 1,001$ (including the intercept). Runtime is reported in seconds. “Iteration” denotes the number of outer iterations, “Rel. grad.” denotes the relative gradient norm at termination, and “Converged” reports whether the solver declared convergence.

Model	Method	Time (s)	Iteration	Rel. grad.	Converged
Gaussian-identity	Spectral-MM	0.050	25	9.38×10^{-9}	Yes
	CRLS	0.033	2	7.38×10^{-9}	Yes
	LSMR	0.033	2	7.38×10^{-9}	Yes
	CGLS	0.034	2	7.34×10^{-9}	Yes
	LSQR	0.035	2	7.34×10^{-9}	Yes
	CG	0.049	2	7.34×10^{-9}	Yes
	<code>glm.fit</code>	2.061	2	1.68×10^{-15}	Yes
Bernoulli-logit [‡]	Spectral-MM	0.217	54	6.21×10^{-9}	Yes
	CRLS	0.631	11	5.29×10^{-9}	Yes
	LSMR	0.631	11	5.29×10^{-9}	Yes
	CGLS	0.643	11	5.32×10^{-9}	Yes
	LSQR	0.644	11	5.27×10^{-9}	Yes
	CG	0.645	11	5.28×10^{-9}	Yes
	<code>glm.fit</code>	54.113	60	4.03×10^{-14}	Yes
Bernoulli-probit [‡]	Spectral-MM	0.155	37	6.68×10^{-9}	Yes
	LSMR	0.632	11	2.91×10^{-10}	Yes
	CRLS	0.633	11	2.91×10^{-10}	Yes
	CG	0.643	11	2.83×10^{-10}	Yes
	CGLS	0.643	11	2.81×10^{-10}	Yes
	LSQR	0.646	11	2.82×10^{-10}	Yes
	<code>glm.fit</code>	38.219	39	2.99×10^{-16}	Yes
Poisson-log	Spectral-MM	0.109	34	1.04×10^{-8}	Yes [†]
	CRLS	0.138	6	5.66×10^{-10}	Yes
	LSMR	0.140	6	5.66×10^{-10}	Yes
	CG	0.149	6	5.28×10^{-10}	Yes
	CGLS	0.152	6	6.05×10^{-10}	Yes
	LSQR	0.152	6	6.08×10^{-10}	Yes
	<code>glm.fit</code>	11.063	11	1.70×10^{-15}	Yes
Gamma-log	Spectral-MM	0.086	59	2.68×10^{-8}	Yes [†]
	CG	0.707	45	7.50×10^{-9}	Yes
	CRLS	0.709	49	5.41×10^{-9}	Yes
	CGLS	0.719	45	7.50×10^{-9}	Yes
	LSQR	0.722	45	7.50×10^{-9}	Yes
	LSMR	0.723	49	5.41×10^{-9}	Yes
	<code>glm.fit</code>	730.592	1000	2.41×10^{-2}	No
Negative-binomial-log	Spectral-MM	0.080	39	2.39×10^{-8}	Yes [†]
	CRLS	0.446	28	7.34×10^{-9}	Yes
	LSMR	0.450	28	7.34×10^{-9}	Yes
	CG	0.493	28	4.98×10^{-9}	Yes
	CGLS	0.498	28	4.98×10^{-9}	Yes
	LSQR	0.502	28	4.98×10^{-9}	Yes
	<code>glm.fit</code>	17.150	17	1.00×10^{-7}	Yes [†]

[†]The solver declared convergence, but neither $\|g\| \leq 10^{-6}$ nor $\|g\|/(1 + \|g_0\|) \leq 10^{-8}$ was satisfied. [‡]These data exhibit separation/perfect prediction; a small gradient does not establish a finite maximum-likelihood estimate. Spectral-MM and the Krylov methods use the shared C++ core. Native timings are medians of five interleaved rounds after warm-up. GLM timings are medians of three runs from an earlier timing session; cross-session timing variation was observed. Sparse GLM inputs were converted to dense, with conversion included in timing. The negative-binomial shape parameter is fixed at $\theta = 4$. All native/Julia fits use $\rho = 10^{-6}$ and no ridge penalty.

Table 6: Computational comparison for generalized linear models using R with a sparse design matrix, $n = 2,000$ and $p = 1,001$ (including the intercept). The non-intercept design has approximately 20 nonzeros per row (about 2% density). Runtime is reported in seconds. “Iteration” denotes the number of outer iterations, “Rel. grad.” denotes the relative gradient norm at termination, and “Converged” reports whether the solver declared convergence.

Model	Method	Time (s)	Iteration	Rel. grad.	Converged
Gaussian-identity	Spectral-MM	0.005	25	1.19×10^{-8}	Yes [†]
	CRLS	0.002	2	6.81×10^{-9}	Yes
	CG	0.002	2	6.76×10^{-9}	Yes
	CGLS	0.002	2	6.76×10^{-9}	Yes
	LSQR	0.002	2	6.76×10^{-9}	Yes
	LSMR	0.002	2	6.81×10^{-9}	Yes
	<code>glm.fit</code>	2.147	2	1.02×10^{-14}	Yes
Bernoulli-logit [‡]	Spectral-MM	0.053	86	5.12×10^{-9}	Yes
	CRLS	0.103	16	1.34×10^{-9}	Yes
	CG	0.104	16	2.59×10^{-9}	Yes
	CGLS	0.104	16	2.00×10^{-9}	Yes
	LSMR	0.106	16	1.34×10^{-9}	Yes
	LSQR	0.106	16	2.93×10^{-9}	Yes
	<code>glm.fit</code>	98.913	98	1.05×10^{-13}	Yes
Bernoulli-probit [‡]	Spectral-MM	0.052	76	9.07×10^{-9}	Yes
	CG	0.089	14	7.70×10^{-9}	Yes
	CGLS	0.090	14	7.94×10^{-9}	Yes
	LSQR	0.093	14	7.66×10^{-9}	Yes
	CRLS	0.095	16	5.48×10^{-9}	Yes
	LSMR	0.099	16	5.14×10^{-9}	Yes
	<code>glm.fit</code>	40.890	40	1.05×10^{-15}	Yes
Poisson-log	Spectral-MM	0.007	30	5.85×10^{-9}	Yes
	CRLS	0.006	5	9.87×10^{-9}	Yes
	CG	0.006	5	9.91×10^{-9}	Yes
	LSMR	0.006	5	9.87×10^{-9}	Yes
	CGLS	0.006	5	9.91×10^{-9}	Yes
	LSQR	0.006	5	9.91×10^{-9}	Yes
	<code>glm.fit</code>	6.091	6	3.35×10^{-13}	Yes
Gamma-log	Spectral-MM	0.016	78	3.43×10^{-8}	Yes [†]
	CRLS	0.036	38	1.69×10^{-8}	Yes [†]
	LSMR	0.038	38	1.69×10^{-8}	Yes [†]
	CG	0.039	40	9.41×10^{-9}	Yes
	CGLS	0.041	41	5.05×10^{-9}	Yes
	LSQR	0.041	40	9.41×10^{-9}	Yes
	<code>glm.fit</code>	364.979	1000	8.25×10^{-2}	No
Negative-binomial-log	Spectral-MM	0.009	34	1.72×10^{-8}	Yes [†]
	CRLS	0.017	16	1.75×10^{-8}	Yes [†]
	LSMR	0.018	16	1.75×10^{-8}	Yes [†]
	CG	0.018	17	6.16×10^{-9}	Yes
	CGLS	0.019	17	6.16×10^{-9}	Yes
	LSQR	0.019	17	6.16×10^{-9}	Yes
	<code>glm.fit</code>	13.128	13	2.06×10^{-7}	Yes [†]

[†]The solver declared convergence, but neither $\|g\| \leq 10^{-6}$ nor $\|g\|/(1 + \|g_0\|) \leq 10^{-8}$ was satisfied. [‡]These data exhibit separation/perfect prediction; a small gradient does not establish a finite maximum-likelihood estimate. Spectral-MM and the Krylov methods use the shared C++ core. Native timings are medians of five interleaved rounds after warm-up. GLM timings are medians of three runs from an earlier timing session; cross-session timing variation was observed. Sparse GLM inputs were converted to dense, with conversion included in timing. The negative-binomial shape parameter is fixed at $\theta = 4$. All native/Julia fits use $\rho = 10^{-6}$ and no ridge penalty.

Table 7: Computational comparison for non-GLM models using R with a dense design matrix, $n = 2,000$ and $p = 1,001$ (including the intercept). Runtime is reported in seconds. “Iteration” denotes the number of outer iterations, “Rel. grad.” denotes the relative gradient norm at termination, and “Converged” reports whether the solver declared convergence.

Model	Method	Time (s)	Iteration	Rel. grad.	Converged
Smooth quantile	Spectral-MM	0.238	40	2.74×10^{-8}	Yes [†]
	CRLS	1.126	21	7.87×10^{-10}	Yes
	CGLS	1.179	20	7.95×10^{-10}	Yes
	LSQR	1.183	20	8.19×10^{-10}	Yes
	LSMR	1.260	23	6.29×10^{-9}	Yes
	CG	1.727	29	8.31×10^{-10}	Yes
Expective	Spectral-MM	0.062	28	1.15×10^{-8}	Yes [†]
	CRLS	0.084	5	3.47×10^{-10}	Yes
	LSMR	0.084	5	3.47×10^{-10}	Yes
	CG	0.089	5	3.59×10^{-10}	Yes
	LSQR	0.090	5	3.59×10^{-10}	Yes
	CGLS	0.091	5	3.59×10^{-10}	Yes
Pseudo-Huber	Spectral-MM	0.067	30	7.55×10^{-9}	Yes
	CRLS	0.171	9	3.79×10^{-9}	Yes
	LSMR	0.175	9	3.79×10^{-9}	Yes
	CG	0.186	9	3.69×10^{-9}	Yes
	LSQR	0.187	9	3.69×10^{-9}	Yes
	CGLS	0.188	9	3.69×10^{-9}	Yes
Student- t	Spectral-MM	0.077	43	5.28×10^{-8}	Yes [†]
	CRLS	0.356	25	7.14×10^{-9}	Yes
	LSMR	0.362	25	6.20×10^{-9}	Yes
	CG	0.371	24	8.54×10^{-9}	Yes
	CGLS	0.375	24	8.54×10^{-9}	Yes
	LSQR	0.377	24	8.54×10^{-9}	Yes

[†]The solver declared convergence, but neither $\|g\| \leq 10^{-6}$ nor $\|g\|/(1 + \|g_0\|) \leq 10^{-8}$ was satisfied. Spectral-MM and the Krylov methods use the shared C++ core. Native timings are medians of five interleaved rounds after warm-up. Parameters: smooth quantile $(\tau, \epsilon) = (0.25, 0.1)$; expectile $\tau = 0.25$; pseudo-Huber $\delta = 1$; Student- t $(\nu, \sigma) = (4, 1)$. All native/Julia fits use $\rho = 10^{-6}$ and no ridge penalty.

Table 8: Computational comparison for non-GLM models using R with a sparse design matrix, $n = 2,000$ and $p = 1,001$ (including the intercept). The non-intercept design has approximately 20 nonzeros per row (about 2% density). Runtime is reported in seconds. “Iteration” denotes the number of outer iterations, “Rel. grad.” denotes the relative gradient norm at termination, and “Converged” reports whether the solver declared convergence.

Model	Method	Time (s)	Iteration	Rel. grad.	Converged
Smooth quantile	Spectral-MM	0.020	43	9.25×10^{-9}	Yes
	CRLS	0.078	19	1.82×10^{-9}	Yes
	LSMR	0.081	19	1.15×10^{-9}	Yes
	CG	0.088	19	3.10×10^{-9}	Yes
	CGLS	0.088	19	5.30×10^{-9}	Yes
	LSQR	0.090	19	4.68×10^{-9}	Yes
Expectile	Spectral-MM	0.006	25	9.93×10^{-9}	Yes
	CRLS	0.006	5	1.32×10^{-10}	Yes
	LSMR	0.006	5	1.32×10^{-10}	Yes
	CG	0.006	5	1.11×10^{-10}	Yes
	CGLS	0.006	5	1.11×10^{-10}	Yes
	LSQR	0.006	5	1.11×10^{-10}	Yes
Pseudo-Huber	Spectral-MM	0.007	26	6.17×10^{-9}	Yes
	CRLS	0.013	10	4.07×10^{-9}	Yes
	CG	0.014	10	4.04×10^{-9}	Yes
	LSMR	0.014	10	4.07×10^{-9}	Yes
	CGLS	0.014	10	4.04×10^{-9}	Yes
	LSQR	0.014	10	4.04×10^{-9}	Yes
Student- t	Spectral-MM	0.009	42	8.82×10^{-9}	Yes
	CRLS	0.025	25	6.23×10^{-9}	Yes
	CG	0.026	25	7.02×10^{-9}	Yes
	CGLS	0.026	25	7.02×10^{-9}	Yes
	LSMR	0.026	25	6.23×10^{-9}	Yes
	LSQR	0.027	25	7.02×10^{-9}	Yes

[†]The solver declared convergence, but neither $\|g\| \leq 10^{-6}$ nor $\|g\|/(1 + \|g_0\|) \leq 10^{-8}$ was satisfied. Spectral-MM and the Krylov methods use the shared C++ core. Native timings are medians of five interleaved rounds after warm-up. Parameters: smooth quantile $(\tau, \epsilon) = (0.25, 0.1)$; expectile $\tau = 0.25$; pseudo-Huber $\delta = 1$; Student- t $(\nu, \sigma) = (4, 1)$. All native/Julia fits use $\rho = 10^{-6}$ and no ridge penalty.